AR 2EB

ALAMEDA RESEARCH
2: ELECTRIC BOOGALOO
LONG HORIZONS · STRUCTURAL SHIFTS · IMAGINATION

INTERNAL RESEARCH · MEMO · NOT INVESTMENT ADVICE · AI-ASSISTED

Illumina

NASDAQ : ILMN · $42.72B mkt cap · 151M sh · $1.16B cash, $1.99B debt · ~$0.83B net debt; ~151M shares; ~$700M/yr buyback (no dividend); Revenue $4.34B FY25 (~flat 4yrs; Q1'26 +4.8% re-accel); ~23% non-GAAP op margin, ~70% gross, ~$0.9-1.0B FCF; ~3.6x off lows on the clinical/NovaSeq-X turnaround; ~10x EV/sales, ~43x FCF

Mature-Company DCF

22 October 2026 · the qualitative

$282.86

THE CENTRAL QUESTION

Is Illumina's ~3.6x run + ~10x EV/sales pricing a durable clinical/NovaSeq-X re-acceleration — or is flat revenue + low-cost-entrant (Ultima/Element) + Roche erosion the reality?

Illumina trades at ~$43B (≈$283/sh, up ~260% off the lows) — ~10x EV/sales, ~43x FCF on $4.34B revenue (flat for four years, Q1'26 +4.8%), ~70% gross margin, ~$1B FCF. The mature DCF asks whether the NovaSeq-X consumables pull-through + clinical mix (>65% of sequencing consumables, +20% ex-China) re-accelerate the franchise toward a ~26% margin, against low-cost entrants (Ultima/Element $100 genome), Roche's 2026 SBX re-entry, China decline, and NIH cuts. The finding: the rally has priced past the re-acceleration — no scenario clears spot; the base (modest re-accel) sits ~62% below. The TXG mirror.

PROBABILITY-WEIGHTED EXPECTED VALUE

most likely case (base) -62.4% · 100% at or below spot

$101.53

expected fair value today

-64.1% vs spot $282.86

FORWARD COMPOUNDED VALUE

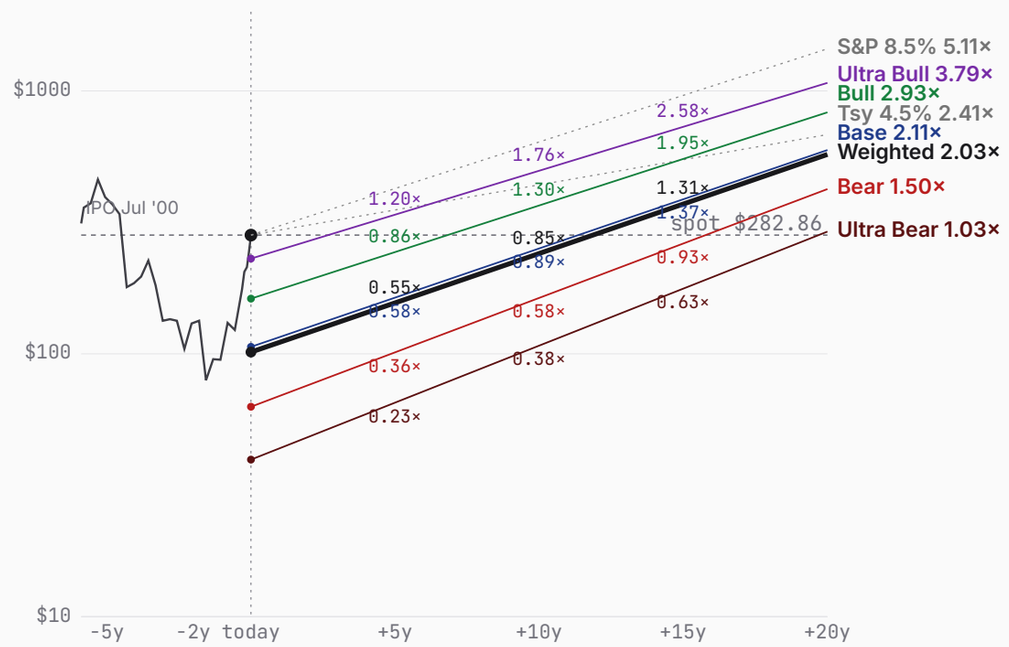
+5y	+10y	+15y	+20y
$156.27	$240.77	$371.34	$573.33
0.55× spot	0.85× spot	1.31× spot	2.03× spot

WEIGHTING RATIONALE

Ultra Bear 12% Entrants + China + NIH crush; ~$40 (-86%).
Bear 30% Flat; clinical offsets; ~$63 (-78%).
Base 35% Modest re-accel; ~26% margin; ~$106 (-62%).
Bull 18% Clinical compounds; multiomics; ~$162 (-43%).
Ultra Bull 5% Sequencing super-cycle; $230 (-19%).

Bull (18%) + Ultra Bull (5%) together contribute $40.70 — 40% of the $101.53 expected value despite 23% combined probability. Today's spot ($282.86) sits 179% above the weighted expected; the forward-weighted value reaches spot in about 12 years.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR — $39.59

-86.0% vs spot

CAGR -1.0% WACC 10.5%
CAGR -1.0% Prob 12%

Low-cost entrants + China + NIH crush.

BEAR — $62.91

-77.8% vs spot

CAGR +1.4% WACC 10.0%
CAGR +1.4% Prob 30%

Flat; clinical offsets erosion; rally unwinds.

BASE — $106.30

-62.4% vs spot

CAGR +4.6% WACC 9.0%
CAGR +4.6% Prob 35%

Modest re-accel; NovaSeq-X pull-through; ~26% margin.

BULL — $162.15

-42.7% vs spot

CAGR +7.2% WACC 8.5%
CAGR +7.2% Prob 18%

Clinical compounds; multiomics; margin >26%.

ULTRA BULL — $230.21

-18.6% vs spot

CAGR +10.0% WACC 8.0%
CAGR +10.0% Prob 5%

Sequencing super-cycle; dominant + multiomics.

NOT INVESTMENT ADVICE · Not from a registered investment advisor · AI-assisted analysis · Author may hold positions · See full disclaimers, page 7

AR 2EB | ALAMEDA RESEARCH 2: ELECTRIC BOOGALOO — LONG HORIZONS · STRUCTURAL SHIFTS · IMAGINATION

Illumina scenario narratives

PROBABILITY WEIGHTS

Ultra Bear 12% Bear 30% Base 35% Bull 18% Ultra Bull 5% · Weighted expected $101.53 (-64.1% vs spot)

ULTRA BEAR $39.59

-86.0% vs spot

Probability 12%

Low-cost entrants + China + NIH crush.

WHY 12%

The bear: the SBS patent estate ages, low-cost entrants design around it, and the academic base shrinks. 12% weight on real erosion.

WHAT HAPPENS

Ultima/Element ($100 genome) compress consumables pricing/share, Roche's SBX re-entry bites, China keeps declining, and NIH cuts freeze research demand. Revenue declines and the consumables annuity erodes.

DCF ~$40 (-86%) — a shrinking franchise losing its pricing Power earns a distressed multiple.

BEAR $62.91

-77.8% vs spot

Probability 30%

Flat; clinical offsets erosion; rally unwinds.

WHY 30%

Flat revenue (the FY22-25 reality) with clinical offsetting erosion. 30% weight — the rally priced more than this.

WHAT HAPPENS

Clinical growth roughly offsets China + entrant erosion, leaving revenue ~flat at a recovering margin; the franchise is dominant but ex-growth.

DCF ~$63 (-78%) — well below the rallied price; flat revenue can't justify ~10x sales.

BASE $106.30

-62.4% vs spot

Probability 35%

Modest re-accel; NovaSeq-X pull-through; ~26% margin.

WHY 35%

NovaSeq-X pull-through + clinical + the ~26% margin target, consistent with Q1'26's +4.8% and raised guide. 35% as the central outcome — but priced in.

WHAT HAPPENS

The modal path: the NovaSeq-X installed base matures into a consumables acceleration, clinical compounds, and the margin marches to ~26% (management's 2027 target). Revenue grows ~4-5%.

DCF ~$106 (-62%) — even a successful modest re-acceleration can't justify the ~260% rally; the entry has run ahead of the fundamentals.

BULL $162.15

-42.7% vs spot

Probability 18%

Clinical compounds; multiomics; margin >26%.

WHY 18%

The clinical-led re-acceleration is durable + multiomics adds growth. ~18%; the rally prices even more than this.

WHAT HAPPENS

Clinical sequencing compounds high-single-digits, multiomics/proteomics (SomaLogic) cross-sells, and the margin pushes past 26% as NovaSeq-X pull-through accelerates in 2H'26+.

DCF ~$162 (-43%) — the durable re-acceleration materializes, yet still well below spot.

ULTRA BULL $230.21

-18.6% vs spot

Probability 5%

Sequencing super-cycle; dominant + multiomics.

WHY 5%

Sequencing super-cycle + ILMN defends its Power. ~5%, and still below spot.

WHAT HAPPENS

A genomics super-cycle (clinical at scale + multiomics) re-accelerates ILMN to low-double-digits at ~30% margins, defending its ~80% share against entrants.

DCF $230 (-19%) — the dominant-platform-re-rates outcome; ~5% probability.

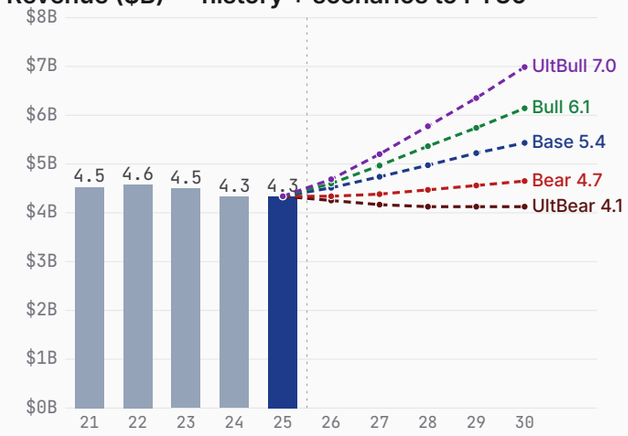
NOT INVESTMENT ADVICE · Not from a registered investment advisor · AI-assisted analysis · Author may hold positions · See full disclaimers, page 7

Illumina business snapshot

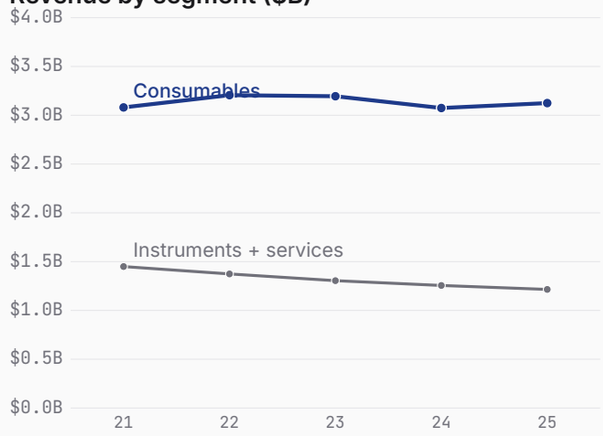
Bear $62.91 Base $106.30 Bull $162.15

FY21-FY25 history + FY26-FY30 scenario projections · calendar fiscal year · FY2025 10-K + Q1'26 (Core Illumina, post-GRAIL)

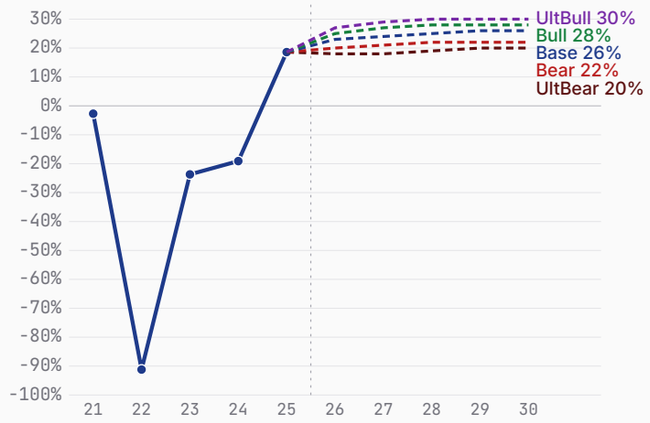
Revenue ($B) — history + scenarios to FY30



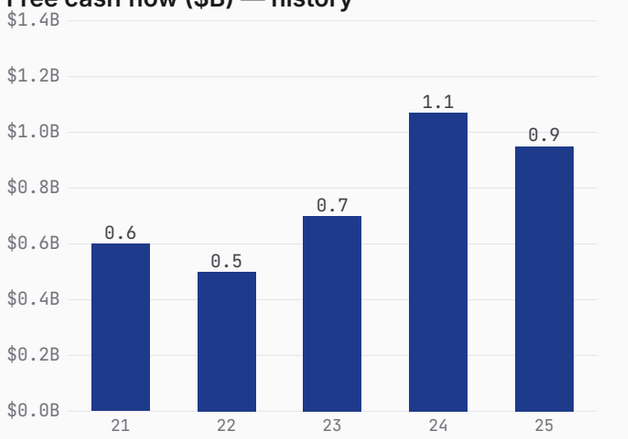
Revenue by segment ($B)



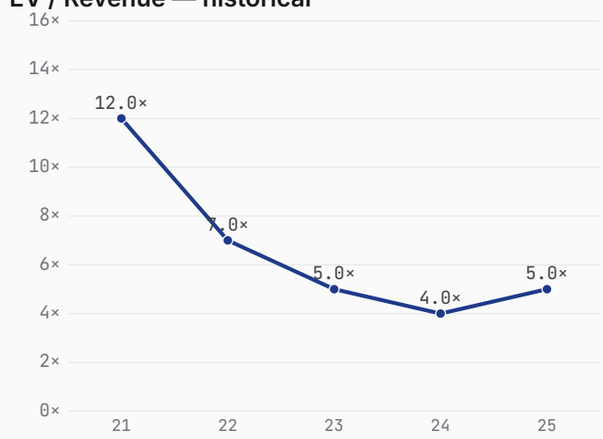
Op margin — history + scenarios



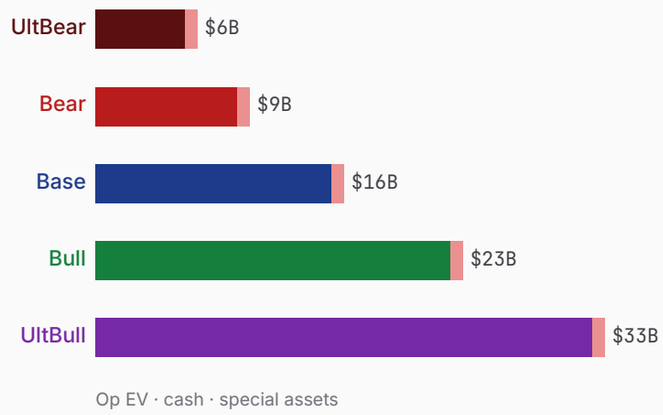
Free cash flow ($B) — history



EV / Revenue — historical



Equity build (Op EV - net debt)



Op EV · cash · special assets

Sources: Illumina FY2025 results (Core revenue $4.34B, ~23% non-GAAP op margin, ~$1B FCF, net debt ~$0.83B), Q1'26 (+4.8%, raised guide). GRAIL divested Jun 2024 → Core-only. Revenue as a growth-rate path off FY25; FCF = revenue x FCF margin; Gordon terminal. EV/sales vs life-science-tools peers.

NOT INVESTMENT ADVICE · Not from a registered investment advisor · AI-assisted analysis · Author may hold positions · See full disclaimers, page 7

Illumina 7 Powers · the moat, scored

Bear $62.91 Base $106.30 Bull $162.15

Arena. DNA sequencing — Illumina (~80% share, NovaSeq-X installed base, ~70%-margin consumables annuity) vs Ultima, Element, Roche (SBX, 2026), PacBio, Thermo, MGI/Complete Genomics; defending a scale + switching Power against low-cost entrants. (Supplies 10x Genomics/TXG; the two are IP litigants.)

POWER AUDIT
dominant-power durability: medium

POWER AUDIT · 7 POWERS

Scale Economies ◂ thesis leans here	~80% share + huge installed base → manufacturing/R&D cost advantage on consumables; the dominant Power.
Network Economies	Published-protocol/citation network is mild.
Counter-Positioning	Incumbent; low-cost entrants counter-position on price-per-genome.
Switching Costs	Validated clinical assays + DRAGEN pipelines on Illumina chemistry won't re-validate cheaply (regulatory friction).
Branding	The 'gold standard' in clinical NGS.
Cornered Resource	Sequencing-by-synthesis patent estate — established but contested/aging (litigated vs Element/BGI).
Process Power	Chemistry/manufacturing learning curve.

PEER SET

Ultima Genomics · private
UG 100, ~$80-100 genome (high-throughput) — the low-cost price attack.

Element Biosciences · private
AVITI + Vitari ($100 30x genome); mutual IP litigation with Illumina.

Roche (SBX) · diversified
Re-entering with sequencing-by-expansion (2026); diagnostics channel — the credible long-term threat.

PacBio / Thermo / MGI · varies
Long-read niche / Ion Torrent / China (MGI blocked from US by BIOSECURE).

THREAT VECTORS · FALSIFIERS

Scale Economies · Ultima / Element ($100 genome)
falsifier: Low-cost entrants take material high-throughput share; consumables pricing compresses.

Cornered Resource · Roche SBX (2026)
falsifier: Roche's SBX gains clinical traction, breaking the SBS franchise.

Switching Costs · NIH / China
falsifier: NIH cuts + China decline shrink the academic installed base the annuity depends on.

COMPETITIVE READ

Illumina's scale + switching-cost Power is genuine and durable — ~80% share, >90% of clinical genomics, a ~70%-margin consumables annuity, and clinical labs that won't cheaply re-validate. But the Power is being audited: revenue was flat for four years, China collapsed, NIH froze research demand, low-cost entrants (Ultima/Element $100 genome) attack pricing, and Roche's SBX re-enters in 2026. The ~260% rally to ~10x EV/sales has priced past the NovaSeq-X/clinical re-acceleration; no scenario clears spot, and the modal base sits ~62% below — the entry, not the franchise, is the problem (the TXG mirror).

NOT INVESTMENT ADVICE · Not from a registered investment advisor · AI-assisted analysis · Author may hold positions · See full disclaimers, page 7

Illumina show your work

Bear $62.91 Base $106.30 Bull $162.15 · Weighted $101.53 (-64.1%)

ULTRA BEAR $39.59

ASSUMPTIONS

5y revenue CAGR (%)	-1.0
Year-5 op margin (%)	20.0
WACC (%)	10.5
Terminal growth (%)	1.0
Terminal FCF ($B)	$0.66B
Starting revenue ($B)	4.34
Scenario probability (%)	12

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	4.25	+18%	+0.68	+0.61
FY27	4.17	+18%	+0.63	+0.51
FY28	4.13	+19%	+0.62	+0.46
FY29	4.13	+20%	+0.66	+0.44
FY30	4.13	+20%	+0.66	+0.40

Σ PV explicit FCF	**+2.43**
+ PV terminal (g 1.0%)	4.26

EQUITY BUILD · $B & PER SHARE

Operating EV	$6.69B
+ Cash & investments	$1.16B
− Total debt	$1.99B
= Equity value	**$5.86B**
FY30 shares (M)	148
= Expected per share	**$39.59**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$65.22	$107.45	$177.02	$291.63
0.23×	0.38×	0.63×	1.03×

BEAR $62.91

ASSUMPTIONS

5y revenue CAGR (%)	+1.4
Year-5 op margin (%)	22.0
WACC (%)	10.0
Terminal growth (%)	2.0
Terminal FCF ($B)	$0.88B
Starting revenue ($B)	4.34
Scenario probability (%)	30

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	4.34	+20%	+0.78	+0.71
FY27	4.38	+21%	+0.79	+0.65
FY28	4.47	+22%	+0.85	+0.64
FY29	4.56	+22%	+0.87	+0.59
FY30	4.65	+22%	+0.88	+0.55

Σ PV explicit FCF	**+3.14**
+ PV terminal (g 2.0%)	7.00

EQUITY BUILD · $B & PER SHARE

Operating EV	$10.14B
+ Cash & investments	$1.16B
− Total debt	$1.99B
= Equity value	**$9.31B**
FY30 shares (M)	148
= Expected per share	**$62.91**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$101.32	$163.17	$262.79	$423.23
0.36×	0.58×	0.93×	1.50×

BASE $106.30

ASSUMPTIONS

5y revenue CAGR (%)	+4.6
Year-5 op margin (%)	26.0
WACC (%)	9.0
Terminal growth (%)	2.5
Terminal FCF ($B)	$1.20B
Starting revenue ($B)	4.34
Scenario probability (%)	35

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	4.51	+23%	+0.90	+0.83
FY27	4.74	+24%	+0.99	+0.84
FY28	4.98	+25%	+1.09	+0.85
FY29	5.23	+26%	+1.15	+0.81
FY30	5.43	+26%	+1.20	+0.78

Σ PV explicit FCF	**+4.10**
+ PV terminal (g 2.5%)	12.25

EQUITY BUILD · $B & PER SHARE

Operating EV	$16.35B
+ Cash & investments	$1.16B
− Total debt	$1.99B
= Equity value	**$15.52B**
FY30 shares (M)	146
= Expected per share	**$106.30**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$163.56	$251.65	$387.20	$595.75
0.58×	0.89×	1.37×	2.11×

BULL $162.15

ASSUMPTIONS

5y revenue CAGR (%)	+7.2
Year-5 op margin (%)	28.0
WACC (%)	8.5
Terminal growth (%)	3.0
Terminal FCF ($B)	$1.54B
Starting revenue ($B)	4.34
Scenario probability (%)	18

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	4.60	+25%	+1.01	+0.93
FY27	4.97	+27%	+1.19	+1.01
FY28	5.37	+28%	+1.34	+1.05
FY29	5.74	+28%	+1.44	+1.03
FY30	6.14	+28%	+1.54	+1.02

Σ PV explicit FCF	**+5.05**
+ PV terminal (g 3.0%)	19.13

EQUITY BUILD · $B & PER SHARE

Operating EV	$24.18B
+ Cash & investments	$1.16B
− Total debt	$1.99B
= Equity value	**$23.35B**
FY30 shares (M)	144
= Expected per share	**$162.15**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$243.82	$366.62	$551.27	$828.92
0.86×	1.30×	1.95×	2.93×

ULTRA BULL $230.21

ASSUMPTIONS

5y revenue CAGR (%)	+10.0
Year-5 op margin (%)	30.0
WACC (%)	8.0
Terminal growth (%)	3.0
Terminal FCF ($B)	$1.96B
Starting revenue ($B)	4.34
Scenario probability (%)	5

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	4.69	+27%	+1.13	+1.04
FY27	5.20	+29%	+1.35	+1.16
FY28	5.78	+30%	+1.56	+1.24
FY29	6.35	+30%	+1.78	+1.31
FY30	6.99	+30%	+1.96	+1.33

Σ PV explicit FCF	**+6.08**
+ PV terminal (g 3.0%)	27.44

EQUITY BUILD · $B & PER SHARE

Operating EV	$33.52B
+ Cash & investments	$1.16B
− Total debt	$1.99B
= Equity value	**$32.69B**
FY30 shares (M)	142
= Expected per share	**$230.21**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$338.25	$497.01	$730.27	$1,073.00
1.20×	1.76×	2.58×	3.79×

NOT INVESTMENT ADVICE · Not from a registered investment advisor · AI-assisted analysis · Author may hold positions · See full disclaimers, page 7

Illumina People · Opportunity · Context · Deal

Bear $62.91 Base $106.30 Bull $162.15

Underwrite the position the way a venture investor underwrites a round — across all four legs, public or private. **People** is the leg scored here (observable only); Opportunity, Context and Deal cross-reference the rest of the memo.

PEOPLE · WHO RUNS / OWNS / FUNDS IT

Jacob Thaysen (CEO since Sept 2023)

2%
insider ownership

LOW
key-person risk

Capital allocation (realized)
Dominated by the GRAIL episode — a major observable value-destruction: bought GRAIL (~$8B, 2021) and CLOSED it without regulatory clearance, drawing an EU prohibition + an unprecedented €432M "gun-jumping" fine (later voided on jurisdiction, 2024) and a forced spin-off (June 2024 → GRAL) that took a ~$4B impairment + a ~$1B recapitalization. No dividend; small buybacks ($1.5B authorized Apr 2026); R&D-heavy.

Incentive alignment
Comp redesigned after a pay revolt: CEO target equity $10M is 70% PSUs / 30% RSUs, PSUs vesting on 3-year relative TSR (the make-whole grant made explicitly 70% TSR-contingent in response to stockholder feedback). Base $1M, target bonus 125%.

GOVERNANCE FLAGS

- single-class, declassified board (annual elections)
- 2023 Icahn proxy fight: chairman John Thompson ousted; Icahn won one board seat
- say-on-pay FAILED 2023 → recovered ~86.5% (2024) after the comp redesign
- board/chair overhaul since (Gottlieb non-exec chair 2025; Corvex Meister joined; Icahn fully exited)

THE FOUR LEGS

People
observable composite · 1–5

Opportunity
The scenario distribution · the business snapshot · the Arthur Indicator (ILMN 13.6)

Context
The 7 Powers analysis (Power Audit)

Deal
Open-market common — entry price weighed against the probability-weighted fair value, then position-sized.
-64.1% vs spot $282.86

> **PEOPLE READ**
> The cohort's governance-turnaround story — and the weakest recent record: a board that closed an $8B deal without clearance (forced unwind, ~$4B impairment, an EU fine), failed say-on-pay (2023), and lost a proxy seat to Icahn. Being repaired — new CEO, independent chair, redesigned TSR-linked pay, say-on-pay recovered, activist oversight — but the track record is the observable flag.

Illumina supporting analysis · disclaimers · glossary

Bear $62.91 Base $106.30 Bull $162.15

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

1 **~80% share, >90% of clinical genomics.**
Dominant installed base + consumables lock-in (validated clinical assays don't re-validate cheaply) — a real scale + switching Power.

2 **~70% gross margin, ~$1B FCF.**
A razor/blade annuity (consumables ~72% of revenue) with a clear path to a ~26% operating margin (2027 target).

3 **NovaSeq-X pull-through is coming.**
~890 installed (end-FY25); ~6-month pull-through lag → a 2H'26 consumables acceleration the base is just starting to price.

4 **Clinical mix is the durable engine.**
>65% of sequencing consumables, +20% ex-China — less funding-exposed than the academic/research side.

5 **GRAIL overhang is gone.**
Divested Jun 2024; Core-only reporting, EU fine/impairments behind it; new CEO running the cost-out + clinical pivot.

FALSIFICATION TRIGGERS

Bull validation
total revenue growth sustains >6% · NovaSeq-X consumables accelerate (2H'26) · non-GAAP op margin reaches ~26% · clinical stays +20% ex-China

Bear validation
revenue stays flat-to-down · Ultima/Element take material share · Roche SBX gains clinical traction · China/NIH drag deepens

Reframe needed
if a low-cost entrant or Roche SBX structurally breaks the SBS consumables-pricing Power, re-rate toward a commoditized-sequencing multiple

DISCLAIMERS · PLEASE READ BEFORE USING THIS DOCUMENT

Not investment advice.

This document is produced for entertainment and educational purposes only by an individual amateur investor. "Alameda Research 2: Electric Boogaloo" is the unincorporated personal concept of one person and is not a registered investment management entity. Nothing herein constitutes investment advice, a recommendation, or a solicitation to buy or sell any security.

Not a professional.

The author is not a registered investment advisor, broker-dealer, CFA charterholder, certified financial planner, or licensed financial professional of any kind. The author has no formal training in equity research or capital markets. No fiduciary relationship is created by reading this document.

AI-assisted analysis.

Substantial portions of this analysis were produced with the assistance of large language models. LLMs are known to fabricate facts, miscalculate numbers, hallucinate sources, and present incorrect information with high apparent confidence. Every figure, claim, and projection in this document should be independently verified before acting on it.

Forward-looking statements.

Scenarios, valuations, projections, and any forward-looking statements involve substantial assumptions and uncertainty. Actual results may differ materially from those projected. Past performance is not indicative of future results. The model is a model; reality is not.

Author may hold positions.

The author may own, intend to acquire, or hold short exposure to $ILMN or related securities at any time. Positions may change without notice and without an update to this document. Do not assume the author's positions match the tone of the analysis.

Do your own research.

Do not make investment decisions on the basis of this document. Consult a licensed financial professional and read the company's official SEC filings (10-K, 10-Q, 8-K, proxy) before forming any investment view. If you wouldn't trust your retirement to a chatbot, don't trust this either.

GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Consumables (razor/blade) — Reagents/flow-cells that run on the installed base (~72% of revenue, ~70% margin) — the recurring annuity and the moat.

Low-cost entrants — Ultima (~$80-100 genome) + Element (Vitari) + Roche SBX (2026) attacking the sequencing-by-synthesis franchise on price.

NovaSeq X — Illumina's flagship high-throughput sequencer; its installed-base ramp drives the consumables pull-through (the re-acceleration mechanism).

FCF margin — Free cash flow / revenue (~21% FY25); the mature-DCF lever as the margin marches to the ~26% 2027 target.

Clinical mix — >65% of sequencing consumables, +20% ex-China — the durable, less-funding-exposed demand engine vs academic/research.